

Moonmoon Mohanty

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Professional Summary

PhD scholar in Electrical and Communication Engineering at the Indian Institute of Science (IISc), specializing in performance analysis, scheduling, and resource allocation for computing and communication systems. Research spans LLM inference optimization, heterogeneous parallel systems, energy minimization, and data-center networks. Hands-on experience modifying vLLM scheduler internals, conducting GPU-based LLM inference experiments, and translating analytical models into deployable scheduling policies.

Technical Skills

Languages	Python, MATLAB
Tools	Linux, Git/GitHub, L ^A T _E X
ML Systems	vLLM
Methods	Queueing theory, stochastic processes, Markov models, performance analysis, scheduling, optimization, discrete event simulation, experimental evaluation
Systems	LLM inference, distributed and parallel systems, heterogeneous computing, data-center networking, Cisco routing and switching

Research and Development Experience

PhD Researcher, Distributed Systems Lab <i>Indian Institute of Science (IISc)</i>	2021–Present <i>Bengaluru, India</i>
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- Developed queueing and stochastic performance models for LLM inference scheduling, characterized system stability and throughput, and identified effective scheduling thresholds under high load.
- Extended the vLLM scheduler to implement and evaluate deferred-prefill scheduling policy; conducted GPU experiments across workload and batching configurations, measured throughput and latency, and validated analytical performance models against system results.
- Contributed to the design, optimization, and experimental evaluation of an energy-aware workload-splitting and CPU-frequency selection framework for heterogeneous cores, achieving up to an 80% reduction in energy-delay product over default Linux governors.
- Analyzed fork-join scheduling on heterogeneous parallel servers, deriving asymptotic workload independence and a near-optimal server-selection policy.
- Investigated matching-based scheduling and resource allocation for data-center networks, focusing on throughput, latency, and scalability; developed a one-round algorithm that outperformed existing approaches in the static regime.

Graduate Researcher, WINNER Lab <i>Ajou University</i>	2015–2017 <i>Suwon, South Korea</i>
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- Designed grouping and fragmentation-based resource-allocation schemes for OFDMA WLANs with heterogeneous packet lengths to improve spectral utilization.
- Modeled MAC and radio-resource-management policies and evaluated them through MATLAB simulations against existing allocation schemes.

Professional Experience

NOC Technician, Level 1 <i>CompuCom, CVS Project</i>	Sep 2013–May 2014 <i>Pune, India</i>
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- Monitored network alarms and service tickets, performed remote fault isolation, and distinguished telecom, power, cabling, and equipment failures.
- Coordinated incident resolution with customer help desks, ISPs, telecom providers, and internal escalation teams while maintaining accurate ticket status and communication.

- Planned and provisioned E1 capacity for new sites, TRX additions, Ater/Gb/X.25 links, and network rollouts; defined end-to-end protection and maintained operational reports.
- Optimized and troubleshot PDH, SDH, microwave, and MUX links; performed error analysis, RSL monitoring, protection testing, commissioning, synchronization, and cross-connection activities.
- Supported GNOC-led rehomeing and upgrades, maintained network inventory in SitePlus, and performed DCN/IP planning, ACL configuration, and Cisco router/switch commissioning.

Education

PhD, Electrical and Communication Engineering <i>Indian Institute of Science (IISc); CGPA: 8.6/10</i> Advisor: Prof. Parimal Parag	2021–Present <i>Bengaluru, India</i>
MS, Electrical and Computer Engineering <i>Ajou University; CGPA: 4.8/5</i> Advisor: Prof. Jae-Hyun Kim	2015–2017 <i>Suwon, South Korea</i>
BTech, Electronics and Communication Engineering <i>C.V. Raman College of Engineering; CGPA: 8.6/10</i>	2005–2009 <i>Odisha, India</i>

Selected Publications and Intellectual Property

C = Conference, J = Journal, S = Preprint, G = Granted Patent

- [S.1] M. Mohanty, G. Bolar, P. Patil, A. Ganesh, J. F. Chamberland, and P. Parag. **Bipartite Matching under Communication Constraints**. *arXiv preprint arXiv:2604.10744*, 2026.
- [C.1] M. Mohanty et al. **Deferred Prefill for Throughput Maximization in LLM Inference**. In *EuroMLSys*, pp. 100–106, 2025.
- [J.1] M. Mohanty, G. Gautam, V. Aggarwal, and P. Parag. **Analysis of Fork-Join Scheduling on Heterogeneous Parallel Servers**. *IEEE/ACM Transactions on Networking*, vol. 32, no. 6, pp. 4798–4809, 2024.
- [C.2] A. Priya et al. **Energy-Minimizing Workload Splitting and Frequency Selection for Guaranteed Performance over Heterogeneous Cores**. In *ACM e-Energy*, pp. 308–322, 2024.
- [C.3] M. Mohanty, J. K. Kim, and J. H. Kim. **Resource Allocation Scheme for Spectral Efficiency Enhancement of OFDMA WLAN Systems**. In *ICEIC*, 2017.
- [G.1] M. Mohanty, J. K. Kim, and J. H. Kim. **OFDMA Wireless LAN System Resource Allocation Method for Higher Spectral Efficiency**. Korea Patent No. 10-1881348.

Awards and Teaching

- CNI Fellowship, IISc, 2022–2023, 2023–2024 and 2025–2026.
- Gold Best Paper Award, IEEE Seoul Section Student Paper Contest, 2016; Best Paper Award, KICS, 2016.
- Graduate Teaching Assistant, E2 202: Random Processes, IISc, Aug–Dec 2022 and E2 202o: Random Processes, Aug–Dec 2025.